

Standard Form — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 (b) 5.9×10^4 . Point moves 4 places; $x = 5.9$ is in range.

Q2 (a) 3.7×10^{-4} . Small number $\rightarrow$ negative power; point moves 4 places right.

Q3 10. x must satisfy $1 \leq x < 10$.

Q4 (b) 308100000. Move the point 8 places right.

Q5 True. Decimals smaller than 1 give a negative power of 10.

Q6 (a) Both true, R explains A. 5.9 is in range and the point moved 3 places.

Short answer

Q7 (i) 3.9×10^8 (ii) 5.678×10^{-6} .

Q8 1.312×10^6 . $3.2 \times 4.1 = 13.12$, and $10^6 \times 10^{-1} = 10^5$; tidy $13.12 \times 10^5 = 1.312 \times 10^6$.

Q9 $(4 \times 10^4) + (7 \times 10^3) + (5 \times 10^2) + (6 \times 10^1) + (1 \times 10^0)$.

Long answer

Q10 (i) 1.496×10^{11} (ii) 1.23×10^{-7} .

Q11 Planet A is nearer. 9.35×10^6 (= 9,350,000 km) is less than 6.27×10^7 (= 62,700,000 km), because $10^6 < 10^7$.

Q12 (a) $2^5 = 32$ cells (b) 2^8 (c) 256. Start 2^3 ; $\times 2^2$ after 2 hours; $\times 2^5$ after 5 hours.

Total: 21 marks.